

A Study on Facial Expression Design Guidelines for Digital Human Modeling - Focusing on Plutchik's Eight Primary Emotions-

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Abstract

This study aims to create a structured framework for designing facial expressions in digital humans, enhancing their emotional realism and user interaction. By employing Plutchik's model of eight primary emotions, the research analyzes the connection between emotions and facial muscle movements, expanding on Paul Ekman's six basic emotions by adding two additional emotional states. Emotions were further categorized by intensity, allowing for more nuanced emotional representations. The study's main contribution lies in developing design principles that go beyond visual replication, fostering emotional interactivity between digital humans and users. These principles enable more natural, emotionally engaging digital interactions, enhancing user immersion in applications requiring emotional intelligence, such as virtual reality and AI-driven platforms. Despite the study's qualitative nature, further research is needed with quantitative data and cross-cultural exploration to strengthen generalizability and ensure the framework's global applicability. Ultimately, this research provides a scientifically grounded approach for improving digital human emotional expressions, with future studies focusing on empirical user evaluations and cross-cultural perspectives to refine its impact across diverse fields.

Keywords: *Digital Human, Facial Expression, Emotion Recognition Model, Plutchik's Wheel of Emotions, Human-Computer Interaction (HCI), Emotional Intensity Classification*

1. INTRODUCTION

Digital humans are advanced 3D models designed to replicate human-like appearance, behavior, and language through digital technologies. With rapid advancements in the metaverse and the growing prevalence of contactless interactions, these virtual entities are increasingly being adopted across various industries, including virtual reality (VR), gaming, education, and healthcare (1). They serve as transformative tools in human-computer interaction (HCI), enabling more dynamic and intuitive engagement. The integration of artificial intelligence (AI) further enhances digital humans by enabling real-time emotional responsiveness, thereby augmenting immersion and fostering deeper user engagement. These developments play a pivotal role in establishing emotional empathy and trust between humans and digital entities (2).

Among various modes of emotional expression, facial expressions serve as the most immediate and powerful form of nonverbal communication. They not only facilitate the direct transmission of emotions but also play a fundamental role in shaping social relationships by reinforcing interpersonal bonds and fostering intimacy. Mehrabian (1981) found that facial expressions account for approximately 55% of perceived communication

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effectiveness, surpassing vocal tone (38%) and verbal content (7%). This finding highlights the critical importance of facial expressions in digital human design, emphasizing that they must extend beyond mere visual representation to function as sophisticated conduits for emotional engagement between humans and virtual agents (3).

Despite their significance, accurately replicating human-like facial expressions in digital humans remains a complex technical and theoretical challenge. Human emotions are inherently nuanced, requiring highly detailed modeling of facial muscle movements to capture subtle variations in affective states. Achieving this level of precision demands an interdisciplinary approach, incorporating insights from psychology, neuroscience, artificial intelligence, and computer graphics. While existing research has proposed various methodologies to enhance the realism of digital human expressions, a comprehensive framework for systematically mapping emotional states onto facial expressions remains insufficient. Without a structured approach, digital humans risk appearing emotionally inconsistent or failing to elicit genuine emotional responses from users.

To bridge this gap, this study seeks to establish a scientifically grounded design framework for digital human facial expressions, drawing upon Robert Plutchik's theory of eight primary emotions. Plutchik's model classifies human emotions into joy, trust, fear, surprise, sadness, disgust, anger, and anticipation, while also explaining how these emotions interact to form more complex affective states (Plutchik, 1980). By applying this framework, the study systematically examines the relationship between emotions and corresponding facial expressions through an extensive review of existing literature and theoretical foundations. Furthermore, expert in-depth interviews were conducted to derive both quantitative and qualitative insights, ensuring high reliability in the proposed design criteria.

The contributions of this study are twofold. First, it advances the technological precision of digital human emotional expressions, providing a structured methodology for refining emotion-driven facial modeling. Second, it lays a theoretical and practical foundation for enhancing human-digital emotional interaction. By developing a systematic and scientifically validated framework, this research contributes to the creation of digital humans that not only visually replicate human emotions but also effectively communicate them in psychologically credible ways. Ultimately, the findings of this study hold significant implications for applications in metaverse environments, virtual reality, AI-driven communication agents, and human-computer interaction, facilitating the development of emotionally intelligent digital entities capable of fostering meaningful engagement with users.

2. Theoretical Background

2.1 Robert Plutchik's Wheel of Emotions

Robert Plutchik defined emotions as psychological mechanisms that have evolved in both humans and animals to adapt to environmental changes. He argued that emotions serve as essential mechanisms for survival and adaptation, playing a critical role in communication and immediate responses to threats. According to Plutchik, an appropriate emotional reaction to external adversities enhances the likelihood of survival, while emotional expressions function as signals that convey intentions for future actions to others. To illustrate this concept, he proposed the "Wheel of Emotions," a multidimensional model that visually represents how emotions are not independent entities, but rather interconnected and opposing forces within a psychological network (4).

According to his theory, human emotions are composed of eight fundamental emotions: joy, trust, fear, surprise, sadness, disgust, anger, and anticipation. Each of these emotions arises for a specific reason and serves an adaptive function. For instance, joy emerges to reinforce the memory of significant experiences, while trust develops to establish connections with those who provide help. Fear functions as a protective mechanism against potential threats, and surprise directs attention to novel stimuli. Sadness plays a crucial role in maintaining emotional bonds with loved ones, whereas disgust serves as a defense against harmful stimuli. Anger motivates individuals to confront and resist challenges, and anticipation enables forward-thinking, allowing for hope, goal-setting, and strategic planning. Furthermore, the process of emotional expression

follows a sequential mechanism consisting of stimulus reception, cognitive appraisal, subjective emotional response, behavioral reaction, and resultant action. This sequence is essential for survival and adaptation, as emotions help individuals navigate their environment effectively. For example, joy fosters social bonding and cooperation through experiences of achievement and satisfaction. Trust reinforces stability and confidence, enhancing interpersonal relationships and collaboration. Fear facilitates rapid escape or defensive actions, thereby improving survival odds. Surprise enhances immediate focus and responsiveness to unexpected events, aiding in the processing of new information. Sadness supports psychological recovery from loss, fostering resilience and social support. Disgust promotes avoidance of harmful substances or situations, ensuring both physical and mental well-being. Anger triggers defensive or offensive responses that drive individuals to challenge injustices and pursue their objectives. Finally, anticipation plays a key role in preparing for future events, enabling individuals to set goals and develop plans to navigate upcoming challenges (5). As shown in Table 1, it becomes evident that emotions manifest in the sequence of 'arousal,' 'cognitive appraisal,' 'subjective response (emotion),' 'behavioral response,' and 'activation.' When applied to specific emotions, this sequence unfolds as follows (6), (7).

Table 1. Eight Primary Emotions and Key Elements of Emotional Sequence

Stimulus events	cognition	Emotions	Overt behavior	Effects	Functional	characteristic	Ego defence	Coping Strategy
Earn something you value	Have	Joy	Repeat	Acquired resources	Reproduction	Sociable	Recoil formation	Invert
Members of your own group	Friends	Trust	Train	Mutual support	Incorporation	Trusted	Negation/Denial	Minimise
Threats	Risk	Fear	Escape	Safety	Protection	Timid	Suppression	Inhibit
Unexpected events	What is it?	Surprise	Stop	Take time to adapt	Adaptation	Indecisive	Regression	Seeking help
Loss of a valued object	Abandonment	Sadness	Cry	Reintegrate with lost audiences	Deprivation	depressed	Rewards	substituted
Offensive targets	Poison	Disgust	Vomit	Expel poison	Rejection	Hostiles	Projection	Finding flaws
Interruption	Enemy	Anger	Attack	Remove obstructions	Destruction	Aggressive	Transposition	Replace
New areas	Investigate	Anticipation	Create a map	Knowledge of the field	Exploration	Demanding	Intellectualization	Mapping

The structural properties of emotions can be explained through three key aspects. First, intensity refers to the variation of emotions from weak to strong states, where the strength of an emotion determines its specific manifestation. For example, a mild form of fear manifests as apprehension, whereas an extreme form of fear escalates to terror. This variation in intensity is visually represented in the Wheel of Emotions model, where the intensity of emotions increases as one moves from the outer to the inner layers. Second, similarity denotes the close positioning of similar emotions, making transitions between them more fluid. For instance, joy and trust are closely related emotions and are placed adjacent to each other due to their high degree of similarity. Third, polarity describes the opposing relationships between emotions, which are positioned on opposite sides of the emotion wheel to illustrate emotional balance. For example, joy is in direct contrast to sadness, trust to disgust, and fear to anger, emphasizing their antagonistic nature (8), (9).

He also explained that emotions can combine to form new complex emotions. For example, the combination of joy and trust results in love, while anger and disgust together produce contempt. Similarly, anticipation and joy merge to create optimism, and disgust and sadness combine to form remorse. These complex emotions are classified into dyads, which involve the blending of two emotions, and triads, which are formed by the combination of three emotions. Through various combinations, new emotional experiences emerge, enriching human affective states and enabling more nuanced and adaptive responses in different situations. Table 2 illustrates the types of emotions that can be experienced through dyadic combinations. Primary dyads arise from the combination of two adjacent basic emotions in the Wheel of Emotions and represent commonly felt emotions. Secondary dyads are formed by combining two emotions that are separated by one petal in the wheel diagram, resulting in emotions that are occasionally experienced. Tertiary dyads involve the combination of

emotions that are separated by two petals in the wheel and are among the least frequently experienced dyadic emotions. Due to the lack of sufficient data, triadic emotions remain unspecified. Through these various emotional combinations, complex emotions are formed, further enriching and refining human emotional experiences, thereby allowing for adaptive responses to a diverse range of situations (10), (11).

Table 2. Robert Plutchik's Mixed Dyadic Emotions

Primary dyad	Joy	Trust	Fear	Surprise	Sadness	Disgust	Anger	Anticipation
Primary dyad	Joy+Trust = Love	Trust+Fear = Obedience	Fear+Surprise = Boundary	Surprise+Sadness = Hostility	Sadness +Disgust = Self-blame	Disgust+Anger = Contempt	Anger+Anticipation = Attack	Anticipation+Joy = Optimism
Secondary dyad	Joy+Fear = Guilty	Trust+Surprise = Curiosity	Fear+Sadness = Despair	Surprise+Disgust = Distrust	Sadness +Anger = Jealousy	Disgust+Anticipation = Cynical	Anger+Joy = Pride	Anticipation+Trust = Hope
Tertiary dyad	Joy+Surprise = Very enjoyable	Trust+Sadness = Sentimental	Fear+Disgust = Shame	Surprise+Anger = Fury	Sadness+Anticipation = Pessimistic	Disgust+Joy = Unhealthy interest	Anger+Trust = Sense of superiority	Anticipation+Fear = Concerned

In conclusion, Plutchik's theory of emotions presents a psycho-evolutionary perspective, emphasizing that emotions are not merely psychological states but essential adaptive mechanisms for survival. By systematically understanding and visualizing emotions, this theory serves as a critical framework for deeply analyzing human emotional experiences and providing practical interpretation and design across various applied fields.

2.2 Anatomical and Functional Roles of Facial Expression Muscles

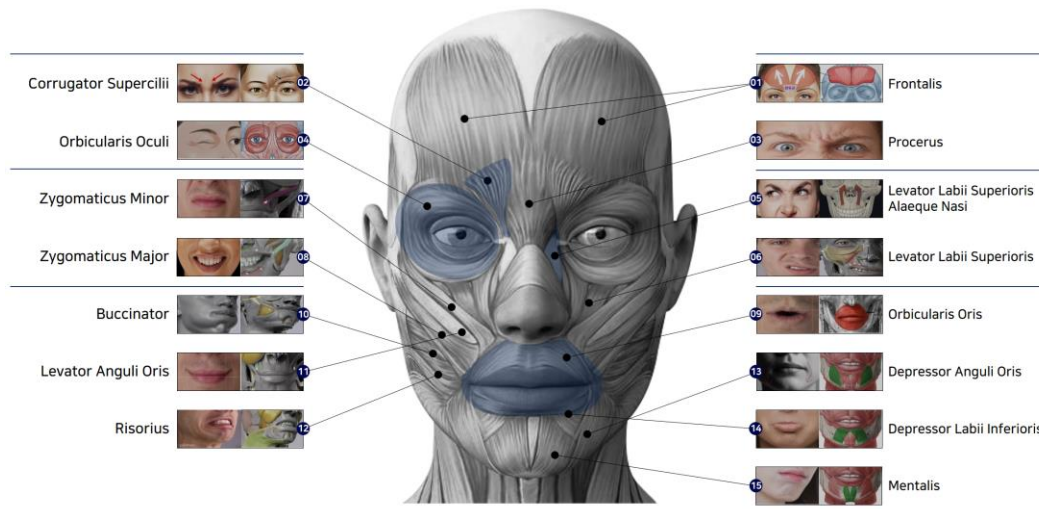


Figure 1.15 Facial Expression Muscles

Facial expression muscles form a highly specialized system responsible for conveying a vast range of human emotions through intricate and nuanced movements. Unlike most skeletal muscles, which attach to bones via tendons, facial muscles connect directly to the skin and subcutaneous tissues, allowing for exceptionally fine motor control essential for nonverbal communication and emotional expression. These muscles can function individually or in coordination, dynamically adapting to emotional states and facilitating complex facial expressions that enhance interpersonal interactions. Given their critical role in human affective communication, a precise understanding of their anatomical structure and functional mechanisms is fundamental for analyzing the mechanisms of emotional response and facial expressivity (12). This study aims to systematically investigate the anatomical positioning and functional properties of facial expression muscles, drawing upon previous empirical research and anatomical studies. Facial expression muscles are categorized into three primary anatomical regions, based on their location and function: superior facial muscles (Musculi Facialis Superior), middle facial muscles (Musculi Facialis Media), and inferior facial muscles (Musculi Facialis Inferior).

Inferior). Each group plays a distinct role in the modulation of facial expressions, allowing for the visual communication of fundamental emotions such as happiness, sadness, anger, fear, and surprise (13). This content pertains to the 15 facial muscles responsible for various expressions and has been meticulously compiled using references from Wiktionary (14), Wikipedia (15), kenhub (16), and namu wiki (17). The detailed descriptions of each muscle's function and role are presented below.

1) Superior Facial Muscles (Forehead and Eyebrows)

The superior facial muscles, which primarily regulate movements of the forehead and eyebrows, play a fundamental role in expressing emotions such as surprise, curiosity, concern, and distress. These muscles include the Frontalis, Corrugator Supercilii, Procerus, and Orbicularis Oculi, each contributing to distinct facial expressions through precise muscular activation.

① **Frontalis (EM1):** The Frontalis is a broad, flat muscle positioned in the forehead, forming the anterior portion of the occipitofrontalis muscle. Unlike most skeletal muscles that attach directly to bone, the frontalis integrates with the epicranial aponeurosis, allowing for fluid movement of the scalp in forward and backward directions. This muscle is primarily responsible for elevating the eyebrows and generating horizontal forehead wrinkles, which play a crucial role in expanding the visual field and modulating responses to external stimuli. It is particularly active in expressions of surprise, curiosity, and confusion, as the elevation of the eyebrows and the deepening of forehead creases amplify the intensity of these emotions. However, prolonged contraction, often associated with chronic stress or persistent anxiety, can lead to deepened forehead wrinkles, reinforcing the visual perception of emotional tension and fatigue.

② **Corrugator Supercilii(EM2):** The Corrugator Supercilii, a small but powerful triangular muscle located at the inner edge of the eyebrows, just above the orbicularis oculi and beneath the frontal muscle fibers, plays a key role in drawing the eyebrows downward and medially. This action creates vertical furrows in the glabellar region, forming the classic frowning expression universally associated with anger, anxiety, worry, and cognitive strain. The corrugator supercilii is instrumental in pain-related facial expressions, accentuating emotional distress by reinforcing a furrowed brow, which serves as a nonverbal cue of discomfort or deep concentration.

③ **Procerus(EM3):** The Procerus, a small triangular muscle extending over the bridge of the nose, functions to pull the glabellar skin downward, leading to the formation of horizontal wrinkles on the upper nasal area. This movement is particularly pronounced in expressions of irritation, contempt, displeasure, and disgust, as the characteristic wrinkling of the nasal bridge enhances the intensity of these negative emotions. The procerus also assists in raising the eyebrows during expressions of surprise or emotional tension, often working in synergy with the corrugator supercilii to create an expression of deep concentration or bewilderment. Beyond emotional expression, the procerus plays a critical role in physiological adaptation to external stimuli. When exposed to intense light, the muscle contracts reflexively, pulling the eyebrows downward to reduce glare and shield the visual field, a protective mechanism crucial for vision preservation.

④ **Orbicularis Oculi(EM4):** The Orbicularis Oculi, a circular muscle encircling the orbit of the eye, is structurally divided into three functional components: orbital, palpebral, and lacrimal portions. The orbital portion facilitates forceful eyelid closure, necessary for squinting and winking, while the palpebral portion regulates blinking and gentle eyelid closure, playing a crucial role in eye lubrication and protection. The lacrimal portion contributes to tear drainage and moisture distribution, maintaining corneal health. The orbicularis oculi is highly active in expressions of happiness and contentment, particularly in the formation of crow's feet wrinkles at the outer corners of the eyes—a hallmark feature of the Duchenne smile, which conveys authentic warmth, friendliness, and social bonding. Conversely, when this muscle remains in a sustained state of tension, it produces expressions indicative of suspicion, anxiety, or heightened vigilance, with the eyes appearing narrowed and tense. The orbicularis oculi also serves a critical protective function—when exposed to bright light or sudden stimuli, it reflexively contracts, narrowing the eyelids to reduce excessive light entry and shield the cornea from potential damage.

2) Middle Facial Muscles (Cheeks and Nose)

The middle facial muscles, positioned centrally between the glabella and the philtrum, play a fundamental role in modulating facial expressions associated with both positive and negative emotional states. These muscles regulate movements in the nasal and zygomatic (cheek) regions, facilitating the expression of joy, amusement, and warmth while also contributing to more complex and nuanced emotions such as contempt and disgust. The principal muscles in this category include the levator labii superioris alaeque nasi, levator labii superioris, zygomaticus minor, and zygomaticus major. These muscles function either independently or in synchrony, allowing for subtle yet highly dynamic emotional expressions, thus enhancing nonverbal communication and reinforcing social interaction.

⑤ **Levator Labii Superioris Alaeque Nasi(EM5):** The levator labii superioris alaeque nasi (EM5) is a long, slender muscle originating from the maxilla (upper jawbone) and inserting into the alar portion of the nose and the lateral region of the upper lip. Functionally, this muscle is responsible for elevating the nasal alae, causing the nostrils to expand, while simultaneously lifting the upper lip, thereby deepening the nasolabial folds. This muscular activation produces a distinctive snarl-like expression, which is strongly associated with negative affective states, particularly disgust, repulsion, and contempt. This muscle intensifies negative emotions by visually signaling rejection and aversion, making it one of the primary contributors to expressions of distaste. It is reflexively activated in response to unpleasant odors or noxious stimuli, reinforcing its role as an innate nonverbal indicator of displeasure. Additionally, the levator labii superioris alaeque nasi enhances aggressive or defensive facial cues by lifting or curling the upper lip, thus strengthening expressions linked to sarcasm, disdain, or mockery.

⑥ **Levator Labii Superioris(EM6):** The levator labii superioris (EM6) is another key muscle in the midfacial region, originating from the maxilla and zygomatic bone and inserting into the skin of the upper lip. Its primary function is to elevate the central portion of the upper lip and draw it laterally, which plays a significant role in the expression of dislike, disgust, scorn, and contempt. This muscle also contributes to the formation of the nasolabial fold, which lifts diagonally to produce a smirking expression. Positioned between the zygomaticus minor and the levator labii superioris alaeque nasi, it allows for fine-grained modulation of facial expressions. The intensity of its contraction directly influences the clarity and impact of the displayed emotion—stronger activations cause a more pronounced elevation of the upper lip and deeper wrinkles around the nose, thereby reinforcing the clarity and emphasis of affective signaling.

⑦ **Zygomaticus Minor(EM7):** The zygomaticus minor (EM7) is a thin, elongated muscle that originates from the zygomaticomaxillary suture of the zygomatic bone and extends toward the upper lip, inserting into the muscular tissue of the orbicularis oris. It is primarily responsible for retracting and elevating the upper lip, contributing to both positive and negative expressions, depending on how it interacts with adjacent muscles. Unlike the zygomaticus major, which originates more laterally, the zygomaticus minor is positioned medially and serves as a critical modulator of fine adjustments in facial expressions. Occasionally referred to as the zygomatic head of the levator labii superioris, this muscle pulls the upper lip upward and outward, forming the characteristic movement of the oral commissures seen in smiles. The zygomaticus minor functions in tandem with the zygomaticus major, adjusting the curvature and intensity of the smile, and influencing expressions of joy, smugness, or sarcasm. Greater activation of the zygomaticus minor leads to a more pronounced smile, whereas asymmetrical contraction can result in expressions associated with cynicism, irony, or contempt.

⑧ **Zygomaticus Major(EM8):** The zygomaticus major (EM8), often referred to as the "laughing muscle," is one of the most essential muscles involved in smiling and laughter. It originates from the zygomatic bone and inserts into the corner of the mouth, where it interacts with the orbicularis oris muscle. When activated, the zygomaticus major pulls the oral commissures upward and laterally, producing a natural and engaging smile that enhances social bonding and interpersonal connection. This muscle works in synergy with the zygomaticus minor and the levator labii superioris, refining both the intensity and emotional nuance of smiling expressions. A genuine Duchenne smile—widely recognized as a marker of authentic happiness—is characterized by the symmetrical contraction of the zygomaticus major, which results in a warm and natural facial expression. However, when asymmetrically activated, this muscle can create a smirk or a cynical expression, in which one side of the mouth lifts higher than the other. This asymmetry contributes to facial cues associated with sarcasm, irony, or mockery, making the zygomaticus major an essential component in the modulation of subtle social and emotional expressions.

3) Inferior Facial Muscles (Mouth and Chin)

The lower facial muscles are integral to controlling movements of the mouth and jaw, spanning from the philtrum (the area beneath the nose) to the chin. These muscles not only play a fundamental role in the expression of negative emotions such as sadness, anger, and disappointment, but they also contribute significantly to essential physiological functions, including chewing (mastication), swallowing, articulation, and emotional communication. The primary muscles in this region include the orbicularis oris, buccinator, levator anguli oris, risorius, depressor anguli oris, depressor labii inferioris, and mentalis. Each of these muscles allows for a wide spectrum of facial expressions and nuanced nonverbal communication, while also supporting key mechanisms related to speech production and food intake.

⑨ **Orbicularis Oris(EM9):** The orbicularis oris (EM9) is a circular muscle that surrounds the lips, consisting of multiple layers of muscle fibers arranged in various directions. It plays a critical role in lip closure, pursing, and

protrusion, thereby facilitating speech articulation, facial expression, and food ingestion. This muscle assists in swallowing by preventing food leakage and regulating intraoral pressure. Additionally, its contraction is responsible for lip pursing, which occurs in expressions of contemplation, tension, or surprise. Depending on the degree of muscle activation, the orbicularis oris enables a broad range of emotional expressions, working in synergy with adjacent muscles to refine the intensity and nuance of facial expressions.

⑩ **Buccinator(EM10):** The buccinator (EM10) originates from the alveolar processes of the maxilla and mandible, as well as from the pterygomandibular raphe, and inserts into the orbicularis oris. This thin, rectangular muscle forms the primary structural foundation of the cheeks and has three fixed insertion points within the inner cheek. The buccinator is essential for compressing the cheeks, which enables the expulsion or inhalation of air, a critical function for chewing, whistling, and infant suckling. Upon contraction, this muscle presses the cheeks against the teeth, preventing food entrapment and assisting the tongue in efficiently moving food within the oral cavity. Additionally, the buccinator plays a role in speech articulation, airflow regulation, and even whistling. In terms of facial expressions, a strong contraction of the buccinator hollows the cheeks, creating a flattened facial appearance, commonly associated with expressions of concentration or worry. Conversely, puffing out the cheeks by maintaining intraoral air pressure may evoke an expression of anger or frustration. This muscle also plays a role in sighing or smiling, modulating a range of emotional and communicative facial expressions.

⑪ **Levator Anguli Oris(EM11):** The levator anguli oris (EM11) originates from the canine fossa of the maxilla and inserts into the subcutaneous tissue of the oral commissure and the orbicularis oris. This muscle plays a crucial role in elevating the corners of the mouth, producing a soft and warm facial expression associated with positive emotions. The levator anguli oris works in synergy with the zygomaticus major, contributing to the formation of natural smiles. Variations in smile shape arise based on the degree of interaction between these muscles. When activated independently, this muscle lifts the corners of the mouth vertically, creating a more open and inviting smile. However, asymmetrical activation of this muscle may result in an awkward or distorted facial expression, sometimes associated with sarcasm or contempt. Prolonged activation of this muscle deepens the nasolabial folds, making them more pronounced over time.

⑫ **Risorius(EM12):** The risorius (EM12) originates from the fascia overlying the masseter muscle and extends to the oral commissure, forming a thin, ribbon-like muscle responsible for pulling the corners of the mouth laterally. The risorius exhibits individual variability, with some individuals possessing an asymmetrical or absent risorius. This muscle is crucial in the formation of dimples, earning it the nickname "dimple muscle". By pulling the oral commissures horizontally, the risorius alters the shape of a smile, aiding in the formation of subtle grins or artificial smiles. It also contributes to bright, open smiles, reinforcing a friendly and approachable impression. When the risorius contracts independently, it pulls the mouth corners outward, potentially creating a tense or rigid expression. However, when coactivated with the zygomaticus major, it results in a more natural and engaging smile. Excessive activation of this muscle can produce an exaggerated smile, which may appear forced or insincere, whereas weakened activation may lead to a restrained or unnatural smile.

⑬ **Depressor Anguli Oris(EM13):** The depressor anguli oris (EM13) originates from the mandible and inserts into the corner of the mouth, playing a vital role in pulling the mouth corners downward, thereby forming a frowning expression. This muscle is strongly associated with negative emotions, such as sadness, disappointment, fatigue, disgust, and cynicism. Upon contraction, the corners of the mouth shift downward and outward, creating a visually elongated and heavier lower facial appearance, intensifying the expression of sorrow or discontent.

⑭ **Depressor Labii Inferioris(EM14):** The depressor labii inferioris (EM14) originates from the oblique line of the mandible, the mental symphysis, and the mental foramen, forming a rectangular muscle extending from the mandible to the lower lip. This muscle plays a key role in expressing negative emotions, such as anger, dissatisfaction, sadness, and depression. It functions to pull the lower lip downward, exposing the lower teeth, and when strongly contracted, it protrudes the lower lip, forming a "pouting" expression. Additionally, this muscle is involved in pulling the lower lip outward and downward, creating a pursed-lip appearance, commonly associated with expressions of frustration or sadness. As emotional intensity increases, the contraction of this muscle visibly amplifies expressions of deep disappointment or frustration. If only one depressor labii inferioris muscle is activated, it creates an asymmetrical expression, allowing for subtle variations in emotional display.

⑮ **Mentalis(EM15):** The mentalis (EM15) originates from the incisive fossa of the mandible and connects to the skin and soft tissues of the chin, forming a small, conical muscle. This muscle is responsible for lifting the lower lip, tightening the chin, and protruding the lips. When activated, the mentalis pulls the lower lip forward while simultaneously contracting the chin, resulting in distinct wrinkles in the mental region. Due to this effect, it is commonly referred to as the "pouting muscle". This muscle plays a significant role in conveying emotions such as discontent, doubt, hesitation, and defiance. When strongly contracted, the mentalis muscle produces an intensely

pronounced facial expression, frequently associated with resentment, suppressed crying, or emotional distress.

3. Methods

This study aims to establish a systematic and scientifically grounded framework for designing emotional expressions in digital humans, ensuring both reliability and validity. To achieve this objective, a structured research methodology was employed, integrating theoretical exploration and qualitative research through expert interviews in a sequential and complementary manner. This approach provided a robust foundation for developing accurate and consistent digital human emotional expression models. The initial phase involved an extensive literature review and theoretical analysis to investigate the intricate relationship between facial expressions and emotions. Plutchik’s Emotion Wheel model was utilized to categorize eight fundamental emotions—joy, trust, fear, surprise, sadness, disgust, anger, and anticipation—and to explore their interrelationships. Additionally, Ekman’s Facial Action Coding System (FACS) was incorporated to systematically classify how each emotion manifests through specific facial muscle movements and activations. This theoretical foundation played a pivotal role in structuring a reliable framework for digital human emotional expression.



Figure 2. 8 Facial Expressions

To refine and validate the accuracy of facial expression modeling, expert interviews were conducted to analyze how these eight emotions are represented through facial expressions. These interviews served as a critical qualitative research method, allowing for an in-depth exploration of specialized knowledge in digital human emotion modeling and facial expression analysis. The study involved one-on-one interviews with five experts, each possessing extensive experience in facial expression analysis, emotion modeling, and digital human design. The interview sessions lasted between 43 to 61 minutes, with an average duration of 50 minutes. During the interviews, pre-collected facial expression images from prior research were presented to the experts, who were tasked with identifying the most appropriate representations for digital humans. Their insights were systematically analyzed to validate and refine the emotional expression models, ensuring a high degree of accuracy, consistency, and realism. Table 3 provides an overview of the general characteristics of the experts involved in this study. Each expert possesses extensive experience in the fields of emotion modeling, facial expression analysis, and digital human design. Their specialized knowledge has been instrumental in enhancing the precision and reliability of this research. The expertise of these professionals played a pivotal role in validating and refining the emotional expression models, ensuring a high degree of accuracy, consistency, and realism.

Table 3. Expert General Characteristics

NO.	Name	Nationality	Related fields	Position
1	이OO	Korea	Counselling Psychology	Research director
2	권OO	Korea	Counselling Psychology	Research director

3	김OO	Korea	Visual Design Producer	Representative
4	조OO	Korea	Cognitive Psychology Research	Ph.D
5	홍OO	Korea	Cognitive Psychology Research	Ph.D

4. RESULTS

4.1 Joy

The facial expression of joy functions as a potent nonverbal cue, effectively conveying positive emotions and psychological well-being. This expression emerges from the coordinated activation and relaxation of multiple facial muscles, with variations in muscular engagement modulating the intensity of the emotional display. The degree of muscular involvement determines the authenticity and perceived sincerity of the expression. In the forehead and eyebrow region, the Frontalis (EM1) muscle remains relaxed, resulting in a smooth, unwrinkled forehead, which contributes to an overall sense of ease and openness. Concurrently, the Procerus (EM3) muscle also relaxes, widening the space between the eyebrows, further reinforcing a calm and inviting facial appearance. These features collectively eliminate tension in the upper face, differentiating joy from expressions associated with stress or concern. Around the eyes, the Orbicularis Oculi (EM4) muscle contracts, producing characteristic wrinkles at the outer corners of the eyes, commonly known as crow's feet. This contraction also causes a slight elevation of the lower eyelid, imparting a softened, natural appearance. In some instances, the eyelids may appear partially closed, reinforcing a sense of warmth and emotional sincerity. These attributes are fundamental to the Duchenne smile, a genuine expression of joy characterized by a harmonious interplay of the orbital and zygomatic muscles, which collectively enhance perceived authenticity and emotional engagement. In the midface and oral region, the Zygomaticus Major (EM8) muscle contracts, lifting the cheeks and deepening the nasolabial folds. Simultaneously, the Levator Anguli Oris (EM11) muscle engages, elevating the corners of the mouth. This action often leads to a slight parting of the lips, revealing the upper teeth, a feature commonly associated with spontaneous and intense joy. The coordinated activation of these muscles amplifies the clarity and intensity of the joyful expression. The intensity of joy is primarily determined by the extent of cheek elevation and mouth opening. A highly expressive joy is characterized by pronounced elevation of the cheeks, a widened mouth, and a lowered jaw, which further accentuates the depth of the nasolabial folds. Conversely, a more subdued joyful expression involves a subtle upward movement of the mouth corners, with the lips either closed or only slightly parted. Regarding the eyes, the degree of eyelid closure does not necessarily correlate with the intensity of joy. Instead, the relaxation of the outer eye corners and the direction of gaze play a more crucial role in reflecting the depth of emotional engagement. When combined, these elements make joy one of the most socially significant and universally recognized facial expressions, playing a fundamental role in interpersonal communication and emotional bonding.

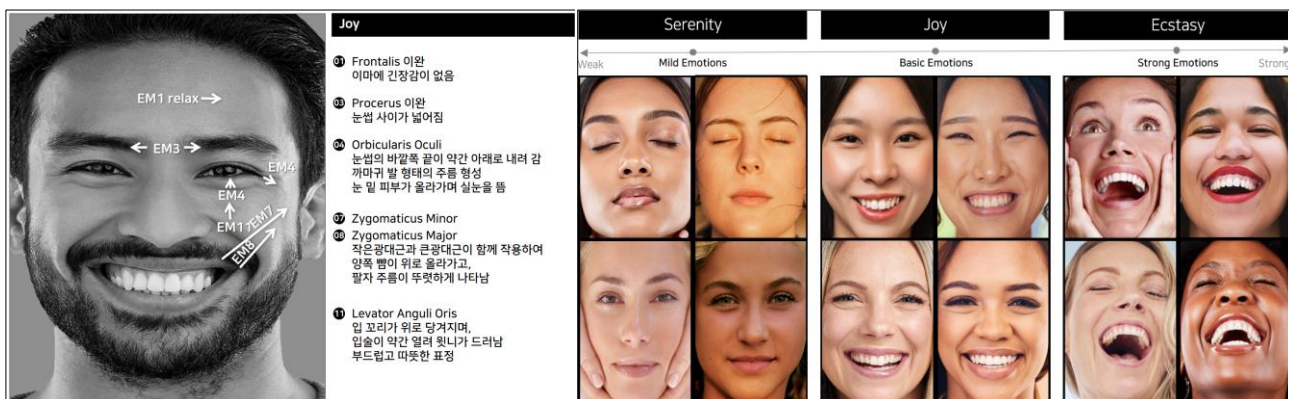


Figure 3. Serenity – Joy – Ecstasy

4.2 Trust

The facial expression of trust serves as a crucial nonverbal indicator of psychological stability, warmth, and openness, signaling a willingness to establish and maintain positive interpersonal relationships. This expression is primarily shaped by the coordinated activation and relaxation of specific facial muscles, with subtle variations that correspond to the intensity of the emotional state. Unlike more overtly expressive emotions such as joy or anger, trust manifests through nuanced facial adjustments that reinforce perceptions of sincerity and attentiveness. In the forehead region, the Frontalis (EM1) muscle remains in a relaxed state, resulting in a smooth, tension-free forehead that contributes to a composed and approachable demeanor. Similarly, the Procerus (EM3) muscle also remains relaxed, preventing the formation of vertical furrows between the eyebrows. This absence of brow contraction minimizes unintentional signals of skepticism or concern, thereby enhancing the expression of psychological ease and receptivity. Around the eyes, the Orbicularis Oculi (EM4) muscle exhibits mild activation, ensuring that the gaze remains steady and engaged. Unlike the expression of joy, in which the outer corners of the eyes contract to produce crow's feet, trust is characterized by an absence of pronounced orbital muscle contraction. Instead, the eyes maintain a naturally open and alert state, enhancing attentiveness toward the interaction partner. This subtle brightening of the eyes functions as a critical visual cue, reinforcing perceptions of sincerity and emotional receptivity. In the midface and oral region, the Zygomaticus Major (EM8) and Levator Anguli Oris (EM11) muscles are mildly engaged, leading to a gentle upward lift at the corners of the mouth. This motion is more subdued than a full smile, often appearing as a slight parting of the lips rather than a broad grin. In some cases, the nasolabial folds or dimples become subtly visible, further enhancing the expression of warmth and openness. This expression often resembles an anticipatory "aha" moment, suggesting an internalized sense of understanding and acceptance. Unlike the exuberant contraction of the zygomatic muscles observed in joy, trust-related expressions remain controlled and composed, avoiding excessive movement that might suggest forced or insincere engagement. The intensity of trust-related emotions varies based on the movement of the eyebrows and mouth. When trust is expressed more strongly, the eyebrows may elevate slightly, forming subtle horizontal lines on the forehead, while the mouth maintains a controlled openness. If the mouth opens wider, the expression begins to blend elements of trust and surprise, reflecting an amplified positive response to new information. This dynamic interaction between cognitive evaluation and affective expression underscores trust as an emotion deeply intertwined with social cognition. Ultimately, the facial expression of trust is not merely a mechanical arrangement of muscle movements but a manifestation of deeper psychological states and neural mechanisms. As a fundamental nonverbal cue in social interactions, trust plays a pivotal role in fostering rapport, conveying sincerity, and enhancing emotional depth. Its expression facilitates smoother interpersonal exchanges, strengthening the foundation for meaningful human-digital interactions, particularly in domains where establishing credibility and engagement is essential, such as virtual assistants, digital human avatars, and AI-driven communication systems.

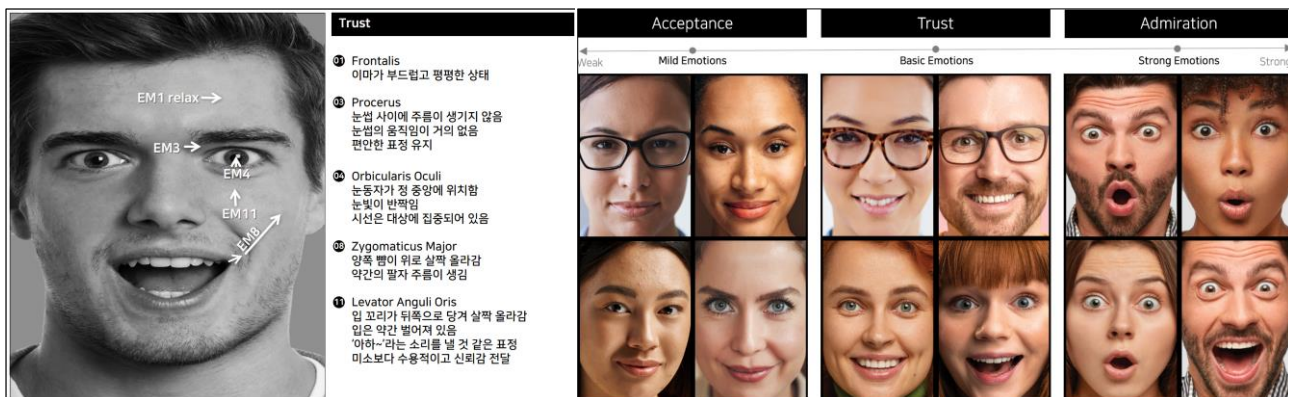


Figure 4. Acceptance – Trust – Admiration

4.3 Fear

The facial expression of fear serves as a powerful nonverbal cue that signals imminent danger and activates an instinctive survival response. This expression is formed through the precise and rapid coordination of multiple facial muscles, facilitating the immediate recognition of threats and the initiation of protective actions. The dynamic interplay of these muscles amplifies the urgency of fear, ensuring its visibility in high-stress situations where rapid response is critical. In the forehead region, the Frontalis (EM1) muscle contracts, producing distinct horizontal wrinkles that extend across the forehead. These lines emphasize the intensity of the emotional state and reinforce the perception of heightened distress. Concurrently, the Corrugator Supercilii (EM2) and Procerus (EM3) muscles engage forcefully, pulling the inner eyebrows sharply upward and inward. This contraction generates deep vertical furrows in the glabella, further intensifying the expression of fear and contributing to a defensive posture. Around the eyes, the Orbicularis Oculi (EM4) muscle activates dramatically, elevating the upper eyelid and widening the eyes to an extreme degree. This action often exposes a large portion of the sclera (white of the eyes), a defining characteristic of fear that enhances visual alertness. Simultaneously, the lower eyelid tenses, increasing ocular rigidity and facilitating rapid scanning of the environment for potential threats. This exaggerated ocular response is an evolutionary adaptation that optimizes visual perception, enabling an immediate assessment of danger and preparation for the fight-or-flight response. In the lower face, the Risorius (EM12) muscle contracts, retracting the corners of the mouth laterally and producing a tense, strained expression. The lips remain rigid, while the jaw drops slightly, keeping the mouth partially open. This positioning allows for rapid breathing and primes the individual for either vocalizing distress (such as a scream) or preparing for immediate physical action. The degree of mouth retraction and jaw separation indicates the severity of fear, with more extreme expressions involving greater muscular tension. The intensity of fear is primarily modulated by the engagement of muscles around the eyes and mouth. Stronger expressions involve a pronounced elevation of the upper eyelid, increased exposure of the sclera, and sharp lifting of the inner eyebrows, which causes the forehead to appear shortened. Additionally, nasal flaring and mouth opening—with teeth remaining clenched—serve to amplify the visual impact of extreme fear. These muscular activations not only communicate distress to others but also prime the body for an immediate protective response. Ultimately, the facial expression of fear is not merely a mechanical reaction but a deeply ingrained neurobiological mechanism designed for survival. Its instantaneous activation functions as an alarm system, signaling potential threats to others and mobilizing defensive actions critical for self-preservation. This innate expression, deeply embedded in human social and evolutionary behavior, ensures that fear remains one of the most universally recognized and compelling emotional displays.



Figure 5. Apprehension – Fear – Terror

4.4 Surprise

The facial expression of surprise emerges instantaneously in response to an unexpected stimulus, serving as a potent nonverbal cue that momentarily halts movement and redirects attention to novel information. This expression is marked by its abrupt onset, heightened intensity, and brief duration, typically lasting no more than one to two seconds before transitioning into another emotional state or naturally fading, depending on the context. The rapid engagement of facial muscles in the forehead, eyes, and mouth occurs in unison, facilitating

the nervous system's swift processing of unanticipated stimuli. In the forehead region, the Frontalis (EM1) muscle contracts, lifting the forehead and creating subtle horizontal wrinkles. This immediate muscular activation reflects the sudden cognitive shift associated with surprise. The eyebrows elevate symmetrically, with the outer edges drawing the surrounding skin into a distinctive arch. Unlike fear, the Procerus (EM3) muscle remains inactive, ensuring that the glabella remains smooth and open, thereby reinforcing an expression of attentiveness rather than distress. Around the eyes, the Orbicularis Oculi (EM4) muscle strongly contracts, elevating the upper eyelid and significantly widening the eyes. This exaggerated eye-opening results in increased exposure of both the superior and inferior sclera, intensifying the expression. The pupils dilate slightly and remain centered, indicating heightened alertness. Unlike in fear, the lower eyelid does not tense but maintains a neutral position, ensuring the absence of defensive tension. In the lower face, the Risorius (EM12) muscle contracts, pulling the corners of the mouth laterally and stretching the lips into a taut, slightly protruded position. Simultaneously, the jaw drops, causing the mouth to open widely. This configuration serves as a visually striking and universally recognized indicator of astonishment, reinforcing the intuitive nature of this response. The intensity of surprise correlates with the degree of mouth opening and jaw displacement. A stronger expression features a more pronounced downward movement of the jaw, a widely open mouth, and maximum elevation of the upper eyelid, further exposing the sclera. In cases of extreme surprise, rapid respiratory shifts often occur, triggering involuntary exclamations such as a "gasp" or an "ah" sound. Physiologically, the expression of surprise is closely linked to autonomic nervous system activation, reflecting the body's rapid adjustment to sudden, unexpected stimuli. Its role as an adaptive response highlights its evolutionary significance, enabling individuals to react swiftly to novel or potentially critical situations. As a fundamental component of social communication, the facial expression of surprise facilitates shared awareness, signaling to others the presence of unexpected information that requires immediate cognitive and behavioral attention.



Figure 6. Distraction – Surprise – Amazement

4.5 Sadness

The facial expression of sadness serves as a powerful nonverbal cue that conveys a profound sense of emotional distress, loss, and an intrinsic need for comfort and empathy. This expression results from the intricate coordination of multiple facial muscles, manifesting naturally as a universally recognized signal of sorrow. The precise engagement of these muscles allows for the nuanced expression of sadness, varying in intensity from subtle melancholy to deep despair. In the forehead region, the Frontalis (EM1) muscle contracts, producing faint horizontal wrinkles that subtly lift the forehead. This upward movement contributes to an overall impression of emotional heaviness across the face, reinforcing the appearance of distress. The Corrugator Supercilii (EM2) muscle activates, drawing the inner portion of the eyebrows upward while causing the outer edges to lower, forming an inverted V-shape. This distinctive brow movement enhances the visual depth of sadness by creating a triangular contour beneath the eyebrows. Additionally, the Procerus (EM3) muscle contracts, intensifying vertical furrows in the glabella, further accentuating the expression of sorrow. In the eye region, the Orbicularis Oculi (EM4) muscle becomes active, leading to a characteristic drooping of

the upper eyelids while the outer corners of the eyes relax. The lower eyelids raise slightly, and the gaze naturally shifts downward, reinforcing a sense of resignation and emotional fatigue. This downward gaze, often associated with sadness, enhances the subdued and introspective nature of the expression, visually amplifying its emotional weight. In the mouth and jaw region, the Depressor Anguli Oris (EM13) muscle pulls the corners of the mouth downward, generating a characteristic frown. Simultaneously, the Depressor Labii Inferioris (EM14) muscle causes the lower lip to protrude slightly, contributing to a subtle pouting effect. This action is further accentuated by the contraction of the Mentalis (EM15) muscle, which elevates the chin, creating fine wrinkles on its surface. These muscular movements collectively shape the distinct appearance of sadness, allowing for an immediate and instinctive recognition of the emotion. The intensity of sadness is determined by the extent of muscle activation. In cases of deeper sorrow, the cheeks lift slightly, and the nasolabial folds (lines extending from the nostrils to the corners of the mouth) become more pronounced. Additionally, the central portion of the lower lip may drop further, the mouth may part slightly, and in extreme emotional states, the lips may tremble. These subtle yet powerful muscular responses amplify the emotional intensity, making sadness one of the most universally understood facial expressions. Ultimately, the facial expression of sadness is not merely a mechanical arrangement of muscle movements but a fundamental component of human emotional communication. It serves as a critical social signal that elicits empathetic responses, fostering connection and support among individuals. By comprehensively understanding the dynamics of this expression, digital human models can be designed to convey sadness with authenticity, thereby enhancing their emotional realism and effectiveness in human-computer interactions.



Figure 7. Pensiveness – Sadness – Grief

4.6 Disgust

The facial expression of disgust serves as a powerful nonverbal cue, signaling an instinctive response to perceived contamination, offense, or repulsion. This expression is closely associated with physiological reactions such as nausea or withdrawal, as it functions to prompt avoidance or expulsion of a harmful stimulus. The distinctive muscular activations involved in disgust vary in intensity, shaping the overall severity and clarity of the emotion being conveyed. In the forehead and eyebrow region, the Corrugator Supercilii (EM2) muscle engages, pulling the inner portions of the eyebrows downward while simultaneously elevating the central portion, creating a characteristic “seagull-wing” shape. This configuration differs from the expression of anger, as the Procerus (EM3) muscle remains inactive, leaving the glabellar region smooth and uncontracted. This absence of vertical furrows in the glabella distinguishes disgust from other negative emotions, such as anger or frustration. Around the eyes, the Orbicularis Oculi (EM4) muscle undergoes mild activation, leading to a subtle lifting of the upper eyelid without excessive tension, maintaining a relatively neutral or slightly narrowed gaze. As the cheeks rise and the lower eyelids relax, faint wrinkles may appear at the outer corners of the eyes (crow’s feet), though eye-related changes play a secondary role in the expression of disgust. Unlike other emotions, the most defining features of this expression manifest prominently in the nasal and oral regions. The Levator Labii Superioris Alaeque Nasi (EM5) muscle plays a critical role in the expression of disgust by forcefully raising the nasal alae and upper lip, resulting in distinct wrinkles along the nasal bridge. This

movement is a key marker of intense aversion and rejection. Additionally, the Levator Labii Superioris (EM6) and Zygomaticus Minor (EM7) muscles engage, elevating one side of the upper lip while simultaneously lifting the cheek, contributing to the overall asymmetry characteristic of this expression. In the mouth and jaw region, the Risorius (EM12) muscle pulls one corner of the mouth laterally and downward, while the opposite side of the upper lip lifts slightly. This asymmetrical distortion of the mouth reinforces the perception of strong disapproval and enhances the overall impact of the expression. The degree of disgust is modulated by the depth of forehead wrinkles and glabellar furrows, as well as the extent of lip retraction and asymmetry. Stronger expressions of disgust feature more pronounced furrowing of the brow, a more exaggerated twisting of the lips, and increased facial asymmetry, effectively intensifying the conveyed repulsion and rejection. Ultimately, the facial expression of disgust is not merely a mechanical response but a deeply ingrained survival mechanism that serves a critical function in human interaction. It acts as an immediate visual warning, allowing individuals to communicate strong disapproval, avoid harmful substances, and reinforce social norms by discouraging undesirable behaviors. By accurately replicating this expression in digital human models, the effectiveness of nonverbal communication in virtual environments can be significantly enhanced, facilitating more immersive and psychologically engaging interactions.

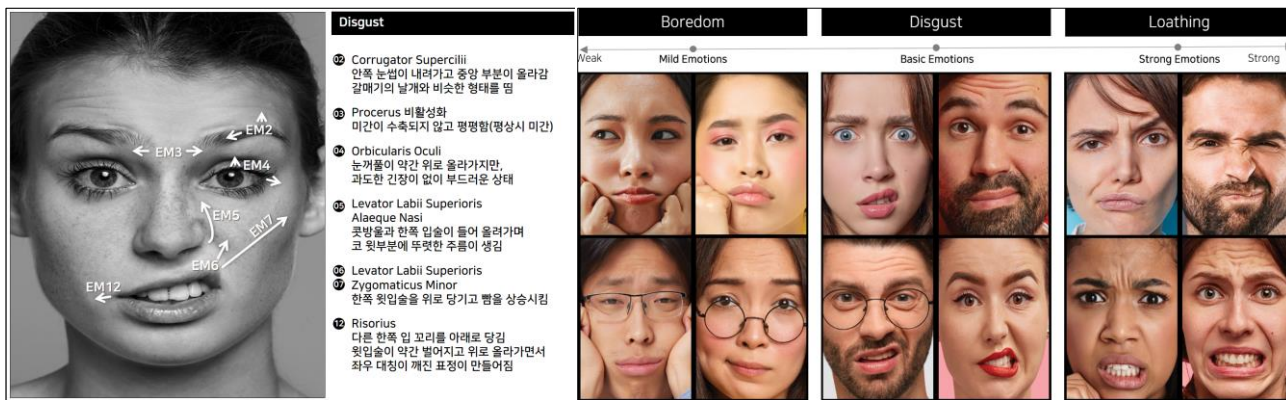


Figure 8. Boredom – Disgust – Loathing

4.7 Anger

The facial expression of anger is a powerful nonverbal signal that conveys intense resistance, frustration, and a strong desire to assert dominance or defend against perceived threats. As an instinctive reaction to external obstruction or coercion, this expression plays a crucial role in human communication by signaling assertiveness, aggression, and readiness for confrontation. The underlying muscle movements involved in anger expression form a structured pattern of activation, reinforcing the emotional intensity and communicative function of this expression. In the forehead and eyebrow region, the Corrugator Supercilii (EM2) muscle contracts forcefully, pulling the eyebrows toward the glabellar region and creating deep vertical furrows between them. Simultaneously, the Procerus (EM3) muscle engages, pulling the inner eyebrows downward toward the nasal bridge, which contributes to the formation of horizontal wrinkles across the upper nose. These muscular contractions intensify the expression of anger, heightening emotional tension and reinforcing the perception of hostility and assertiveness. Around the eyes, the Orbicularis Oculi (EM4) muscle activates, depressing the inner eyebrows while elevating the upper eyelid. This dual action widens the eyes, creating a sharp, focused, and penetrating gaze. The heightened ocular tension and increased scleral exposure amplify the perception of dominance and aggression, establishing a confrontational posture that serves as a warning signal in social interactions. In the nasal and upper lip region, the Levator Labii Superioris Alaeque Nasi (EM5) muscle contracts, lifting the nasal alae and upper lip, which leads to the formation of distinct wrinkles along the nasal bridge. This contraction causes the nostrils to flare and gives the nose a slightly shortened appearance, visually enhancing the impression of irritation, defiance, and assertive readiness. The mouth and cheek region exhibit distinct changes that further reinforce the intensity of anger. The Buccinator (EM10) muscle tenses, ensuring that the lips remain firmly pressed together, maintaining strong intraoral

pressure. This tight-lipped configuration often signifies controlled anger or suppressed aggression. In situations of heightened emotional intensity, the lips may part slightly to reveal clenched teeth, emphasizing a readiness to engage in confrontation. Additionally, activation of the Levator Labii Superioris (EM6) and Depressor Labii Inferioris (EM14) muscles causes the upper lip to lift while the lower lip projects forward, increasing overall facial tension and enhancing the perception of aggression. Further modifications occur in the lower facial region, where the Depressor Anguli Oris (EM13) muscle pulls the corners of the mouth downward, creating a thinner, more compressed lip shape that reinforces a stern, aggressive appearance. At the same time, activation of the Mentalis (EM15) muscle draws the chin downward, producing prominent wrinkles in the chin area, which further intensifies the expression of determination and defiance. The intensity of anger is modulated by the degree of eyebrow contraction, eyelid elevation, and jaw positioning. A more extreme expression involves deeper vertical furrows in the glabellar region, a pronounced opening of the mouth that reveals clenched teeth in a rectangular mouth shape, and a forward projection of the jaw, enhancing the perception of dominance and aggression. Additionally, as the jaw juts forward and facial redness intensifies due to sympathetic nervous system activation, the overall expression of extreme anger becomes more visually striking and socially impactful. This complex interplay of facial muscle activation not only conveys emotional intensity but also serves as an evolved nonverbal mechanism for effectively signaling dominance, resistance, and readiness for confrontation. The ability to accurately interpret and reproduce this expression in digital human models is essential for creating realistic and psychologically convincing virtual interactions, particularly in applications where emotional authenticity and nonverbal communication play a crucial role.



Figure 9. Annoyance – Anger – Rage

4.8 Anticipation

The facial expression of anticipation conveys a forward-looking, exploratory mindset, visually reflecting curiosity and a positive emotional response toward future possibilities. This expression signifies not only hope for a desired outcome but also cognitive preparedness and adaptive readiness for expected events. By subtly integrating muscle movements across the face, anticipation becomes a nuanced nonverbal signal that enhances engagement and social connection. In the eyebrow region, the Procerus (EM3) muscle remains inactive, preventing contraction of the eyebrows and the glabellar region, thereby maintaining a neutral and relaxed appearance. The outer ends of the eyebrows naturally lower, fostering an open and receptive demeanor. This subtle downward movement contributes to a stable expression that enhances the perception of emotional engagement without signs of tension or apprehension. Around the eyes, the Orbicularis Oculi (EM4) muscle is lightly activated, directing the gaze upward in a natural and subtle manner. This gentle muscle engagement causes a slight elevation of the upper eyelid while shifting the pupil upward, occasionally revealing a small portion of the sclera (white of the eye). These features are commonly observed during states of focused thought and cognitive engagement, serving as key visual cues of attentiveness, curiosity, and expectancy. In the cheek and mouth regions, the Zygomaticus Major (EM8) muscle contracts, subtly lifting the cheeks while pulling the corners of the mouth upward. This results in a natural smile that often reveals the upper teeth, reinforcing an emotional state infused with optimism and anticipation. The Levator Anguli Oris (EM11) and Risorius (EM12)

muscles further enhance this expression by contributing to the upward movement of the mouth corners, sometimes leading to the formation of nasolabial folds or dimples. The intensity of these muscular activations varies depending on the level of anticipation, with stronger expressions manifesting as a more pronounced elevation of the cheeks and deeper dimples. The degree of anticipation is modulated by fine-tuned facial muscle adjustments. As anticipation strengthens, the eyebrows slightly rise, forming subtle horizontal wrinkles on the forehead. One corner of the mouth is lifted higher than the other, creating a cautious or evaluative expression. These nuanced variations illustrate anticipation as a complex and dynamic emotional state, bridging curiosity with preparation for future events. Beyond its role as a hopeful emotion, anticipation serves as a fundamental psychological mechanism for goal-directed behavior, enhancing motivation, cognitive engagement, and social interaction. By fine-tuning digital human expressions to capture this emotional state, virtual entities can foster more authentic and immersive interactions, particularly in applications where engagement and predictive responsiveness are essential.



Figure 10. Interest – Anticipation – Vigilance

5. Conclusion

This study aimed to establish a systematic design framework that enables digital humans to accurately and effectively express human emotions. To achieve this goal, Plutchik's eight primary emotions model was used as the theoretical foundation, and the relationship between emotions and facial muscle movements was analyzed in depth. Additionally, expert in-depth interviews were conducted to enhance the reliability and validity of the study, which led to expanding Ekman's original six basic emotions by incorporating two additional emotions. These emotions were further classified into strong and subtle affective states based on their intensity. This structured approach contributed to the development of a framework for digital human emotional expression that ensures greater precision and reliability.

The key contribution of this study lies in its shift beyond mere visual replication of emotions. It establishes design principles that position the emotional expressions of digital humans as a fundamental mechanism for fostering emotional engagement with users. Through improving the accuracy of emotional expressions and enhancing user immersion, the study empirically demonstrates that digital humans can engage in more natural, emotionally resonant interactions with users. These findings underscore the critical importance of digital humans in fields such as virtual reality, artificial intelligence, and interactive media, where emotional communication is essential, and suggest that digital humans could become indispensable in enhancing emotional interaction.

However, this study also presents several limitations. First, as it primarily focused on qualitative analysis, further research is necessary to enhance the generalizability of the findings through additional quantitative data analysis and experimental validation. Particularly, assessing the impact of digital human emotional expressions on user experience using quantitative methodologies will be a key task in future research. Additionally, the study did not fully explore the influence of cultural differences on emotional expression. Since emotional expressions can be interpreted differently across cultural contexts, investigating how digital human emotions are perceived and utilized in diverse cultural settings is essential.

In conclusion, this study provides a scientifically grounded, systematic approach to designing digital human emotional expressions, contributing to a deeper understanding of emotional interactions between humans and digital entities. Future research should focus on empirical evaluations of user experience and in-depth analysis of emotional expressions in different cultural contexts to further optimize digital human affective design. By continuing research and development in this field, the emotional communication abilities of digital humans can be enhanced, maximizing their applicability in global markets and reinforcing their role as essential interactive agents across various digital environments.

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